

Variations of the maximum leaf spanning tree problem for bipartite graphs

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Abstract

The *maximum leaf spanning tree* problem is known to be NP-complete. In [M.S. Rahman, M. Kaykobad, Complexities of some interesting problems on spanning trees, Inform. Process. Lett. 94 (2005) 93–97], a variation on this problem was posed. This variation restricts the problem to bipartite graphs and asks, for a fixed integer K , whether or not the graph contains a spanning tree with at least K leaves in one of the partite sets. We show not only that this problem is NP-complete, but that it remains NP-complete for planar bipartite graphs of maximum degree 4. We also consider a generalization of a related decision problem, which is known to be polynomial-time solvable. We show the problem is still polynomial-time solvable when generalized to weighted graphs.

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1. Introduction

Let $G = (V, E)$ be a connected graph and T a spanning tree of G . A vertex of degree 1 in T is called a *leaf* (node) of the tree T . Consider the following problem.

Problem 1.1 (*Maximum leaf spanning tree*). Let $G = (V, E)$ be a connected graph and let $K < |V|$ be a positive integer. We are asked whether G contains a spanning tree T consisting of at least K vertices of degree 1.

The *maximum leaf spanning tree* problem is known to be NP-complete (see [1]). In fact, it remains NP-

complete for regular graphs of degree 3 (see [3]) as well as for planar graphs of maximum degree 4 (see [1]). The optimization version of this problem is known to be NP-hard. This problem has applications in the area of communication networks (see [5]). For example, a connected graph $G = (V, E)$ may model a computer network where vertices V correspond to the computers and edges E correspond to the computers that may communicate directly with one another. A spanning tree T of G provides a subset of the communication links by which any two computers may communicate, using only this subset of links. The nodes of degree one in the tree correspond to computers that performs less communication processing than nodes of degree at least two. Therefore, it is reasonable to ask for a spanning tree of G with the maximum number of leaf nodes.

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In [4], the following variation on the maximum leaf spanning tree problem was introduced for bipartite graphs.

Problem 1.2 ([4], *variant of maximum leaf spanning tree problem for bipartite graphs*). Given a connected graph $G = (V, E)$ with partite sets X and Y and a positive number $K \leq |X|$, we are asked the question of whether there is a spanning tree T_G of G such that the number of leaves in T_G belonging to X is greater than or equal to K .

The authors were able to reduce this problem to an equivalent problem, as stated by the following result.

Theorem 1.3 [4]. *Let $G = (V, E)$ be a connected bipartite graph with partite sets X and Y . Let $K \leq |X|$ be a positive integer. Then there is a spanning tree T of G with at least K leaves in X if and only if there is a set $S \subseteq X$ such that*

- (1) $|X \setminus S| \geq K$, and
- (2) *the induced subgraph $S \cup Y$ of G is connected.*

Unfortunately, this characterization of Problem 1.2 did not lead to a resolution of its complexity. It was stated by the authors of [4] that it is currently unknown whether Problem 1.2 is NP-complete. We will show that Problem 1.2 is NP-complete using Theorem 1.3. We will also show that this problem remains NP-complete for planar bipartite graphs of maximum degree 4. To the best of our knowledge, these are the first complexity results related to this problem.

2. Main result

We begin by showing that Problem 1.2 is NP-complete. To do this, we will reduce an instance of the set-covering problem to an instance of the problem stated in Theorem 1.3.

Theorem 2.1. *Problem 1.2 is NP-complete.*

Proof. Consider an instance of the set-covering problem given by a collection $\mathcal{C}$ of subsets of the finite set $A = \bigcup_{C \in \mathcal{C}} C$ and a positive integer $c \leq |\mathcal{C}|$. This problem is NP-complete (see [1]). Construct a bipartite graph $G = (X \cup Y, E)$ from the incidence relation between $\mathcal{C}$ and A so that X correspond to $\mathcal{C}$ and Y correspond to A . To ensure that G is connected, add an extra element, ∞ , to the set Y and include it in every set of $\mathcal{C}$. Let $K = |X| - c$.

It is easy to see that any set cover of A induces a connected subgraph in G which contains all of Y . On the other hand, any family of subsets that induces a connected subgraph of G which contains all of Y , is a set cover of A . Therefore, $\mathcal{C}$ contains a cover for A of size c or less if and only if there exists a set $S \subseteq X$ such that $|X \setminus S| \geq K$, and the induced subgraph $S \cup Y$ of G is connected. By Theorem 1.3, the desired result immediately follows. $\square$

We now show that Problem 1.2 remains NP-complete if we consider only bipartite graphs of maximum degree 4. We do this by providing a reduction from the exact cover by 3-sets problem. We now formally state the exact cover by 3-sets problem which is known to be NP-complete (see [1]).

Problem 2.2 (*Exact cover by 3-sets (X3C)*). Given a finite set X with $|X| = 3q$ and a collection $\mathcal{C}$ of 3-element subsets of X , we are asked the question of whether there is a sub-collection of $\mathcal{C}$ that partitions X .

The X3C problem remains NP-complete if no element of X occurs in more than three subsets of $\mathcal{C}$ (see [1]). We now formally state a restricted version of Problem 1.2 and show that it is NP-complete.

Problem 2.3. Given a connected bipartite graph $G = (V, E)$ of maximum degree 4 with partite sets X and Y and a positive number $K \leq |X|$, we are asked the question of whether there is a spanning tree T_G of G such that the number of leaves in T_G belonging to X is greater than or equal to K .

Theorem 2.4. *Problem 2.3 is NP-complete.*

Proof. Let $(X, \mathcal{C})$ be an instance of the X3C problem with $|\mathcal{C}| = p$, $|X| = 3q$, $p \geq q$, and no element of X occurs in more than three members of $\mathcal{C}$. We will now construct a bipartite graph $G = (V, E)$ with maximum degree 4 with partite sets A and B , such that $(X, \mathcal{C})$ has an exact 3-cover if and only if G contains a spanning tree with at least $p - q$ leaves in B . We begin by building a rooted tree T^* (from the bottom up) of depth $\lceil \log_2 p \rceil$ as follows:

- (1) Let the members of $\mathcal{C}$ be the nodes at depth $d = \lceil \log_2 p \rceil$ of the tree and color each node white.
- (2) Partition the nodes at depth d into sets of two nodes until it cannot be done and put the remaining node, if it exists, as a (singleton) member of the partition.

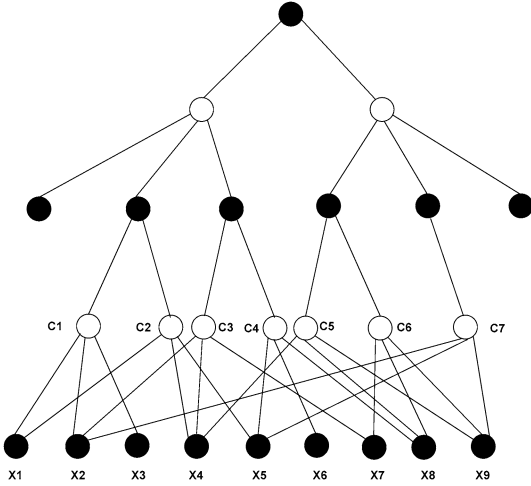


Fig. 1. The graph constructed from an X3C instance.

- (3) For each member of the partition, create a new node v (with color black) and join the nodes in the member to it. The newly created node will be a node at depth $d - 1$. If the node v is at an even distance from a node of $\mathcal{C}$, then
 - (a) re-color v white,
 - (b) adjoin a new node u to v , where the depth of u is $\text{depth}(u) = \text{depth}(v) + 1$,
 - (c) color u black.
- (4) Set $d = d - 1$. If $d = 0$, stop. Otherwise, go to step 2.

This concludes the construction of the tree T^* . Now create G from T^* by connecting the members of $\mathcal{C}$ (in T^*) and X as follows: For $c \in \mathcal{C}$ and $x \in X$, $c \rightarrow x$ if and only if $x \in c$. Finally, color all members of X black. Taking all the black nodes as A , and all the white nodes as B along with all the edges E created by the above process, $G = (A \cup B, E)$ is clearly a bipartite graph with partite sets A and B , and maximum degree 4. It is clear that the graph G can be constructed in polynomial time. Fig. 1 illustrates the graph that is constructed from the X3C instance $(X, \mathcal{C})$ where $X = \{x_1, x_2, \dots, x_9\}$ and $C_1 = \{x_1, x_2, x_3\}$, $C_2 = \{x_1, x_4, x_5\}$, $C_3 = \{x_2, x_4, x_7\}$, $C_4 = \{x_5, x_6, x_8\}$, $C_5 = \{x_4, x_8, x_9\}$, $C_6 = \{x_7, x_8, x_9\}$, and $C_7 = \{x_2, x_5, x_9\}$.

The key observation here is that none of the white nodes (those in B) can be leaf nodes of a spanning tree, unless they are members of $\mathcal{C}$. This is because of the (black) leaf nodes that were introduced in step 3b of the construction. We will show that $(X, \mathcal{C})$ has an exact 3-cover if and only if G contains a spanning tree with at least $K = p - q$ leaves in B . If there is an exact 3 cover say $\{c_1, c_2, \dots, c_q\} \subseteq \mathcal{C}$, then we can choose the span-

ning tree T_G by taking the tree T^* . In addition, for each $c_i = \{x, y, z\}$ in the exact cover, add the edges $\{c_i, x\}$, $\{c_i, y\}$, $\{c_i, z\}$ to T_G . T_G is clearly a spanning tree of G .

Conversely, let T_G be a spanning tree of G such that B contains at least $p - q$ leaf nodes. From our observation that only the nodes in $\mathcal{C}$ may be leaf nodes in B , it follows that at least $p - q$ members of $\mathcal{C}$ are leaf nodes in T_G . It is clear that if we can show that $\mathcal{C}$ contains exactly $p - q$ leaf nodes, we can construct an exact 3-cover. Suppose there are at least $p - q + 1$ members of $\mathcal{C}$ which are leaf nodes in the spanning tree T_G . Then there are at most $q - 1$ members of $\mathcal{C}$ which are non-leaf nodes. These are the only nodes that connect the members of X to G . However, note that $|X| = 3q$ and each member of $\mathcal{C}$ is adjacent to at most 3 members of X in G . Since $(q - 1) * 3 = 3q - 3 < 3q$, this contradicts the assumption that T_G is a spanning tree. Therefore, there are exactly $p - q$ members of $\mathcal{C}$ that are leaf nodes. $\square$

Problem 2.3 remains NP-complete if we assume the bipartite graph is planar. This is due to the following result from [2].

Theorem 2.5. *1-in-3 satisfiability remains NP-complete if*

- (1) *Every variable appears in exactly 3 clauses,*
- (2) *Negations do not occur in any clause, and*
- (3) *The bipartite graph formed by joining a variable and a clause if and only if the variable appears in the clause, is planar.*

It can easily be seen that an instance of the restricted 1-in-3 satisfiability problem, given by the conditions in Theorem 2.5, can be reduced to an instance of the X3C problem $(X, \mathcal{B})$ by associating X with the clauses and $\mathcal{B}$ with the variables of the satisfiability problem instance. Using this result, we can use a planar instance of the X3C problem $(X, \mathcal{B})$ where each element of X occurs in exactly 3 members of $\mathcal{B}$ in the proof of Theorem 2.4 to obtain the desired result. The result of the preceding argument is stated in Theorem 2.6.

Theorem 2.6. *Problem 1.2 remains NP-complete for planar bipartite graphs of maximum degree 4.*

3. Other results

In this section, we generalize a problem that was shown in [4] to be polynomial-time solvable. We will

show that the generalized problem is also polynomial-time solvable. We begin by stating the original problem.

Problem 3.1 [4]. Given a connected graph $G = (V, E)$, and $X \subseteq V$, we are asked whether there is a spanning tree T of G such that $X \subseteq \Pi_T$, where $\Pi_T = \{v: v \text{ is a leaf node of } T\}$.

It was shown in [4] that Problem 3.1 is polynomial-time solvable. Given a weighted graph $G = (V, E)$, each edge $\{x, y\} \in E$ is assigned a positive value, known as the *weight* of the edge. The weight of the edge $\{x, y\}$ is denoted by $wt(\{x, y\})$. If T is a tree, let $wt(T)$ denote the sum of the weights of the edges in the tree T . We now generalize Problem 3.1 for weighted graphs.

Problem 3.2. Given a connected, weighted graph $G = (V, E)$, $X \subseteq V$ and $k > 0$, we are asked whether there is a spanning tree T of G such that $wt(T) \leq k$ and $X \subseteq \Pi_T$, where $\Pi_T = \{v: v \text{ is a leaf node of } T\}$.

The following result characterizes the existence of a spanning tree that satisfies the properties given in Problem 3.2.

Theorem 3.3. Let $G = (V, E)$ be a connected, weighted graph, $X \subseteq V$, $k > 0$, and $Y = V \setminus X$. Then there exists a spanning tree T of G such that $wt(T) \leq k$ and $X \subseteq \Pi_T$ if and only if the following three conditions hold:

- (1) The subgraph Y of G is connected,
- (2) Every vertex of X is adjacent to a vertex of Y , and
- (3) The subgraph Y of G has a spanning tree with weight at most $k - \sum_{x \in X} \min\{wt(\{x, y\}): \{x, y\} \in E, y \in Y\}$.

Proof. Suppose that there exists a spanning tree T of G such that $wt(T) \leq k$ and $X \subseteq \Pi_T$. Then it was shown in [4] that conditions (1) and (2) hold. To see that condition (3) holds, we observe that each vertex $x \in X$ must be a leaf node of T . Hence, the edges in T , involving each $x \in X$, must be of the form $\{x, y\}$ where $y \in Y$. These edges have aggregate weight of at least $\sum_{x \in X} \min\{wt(\{x, y\}): \{x, y\} \in E, y \in Y\}$. Therefore, the subgraph $Y \cap T$, which is a spanning tree of Y , must have weight at most $k - \sum_{x \in X} \min\{wt(\{x, y\}): \{x, y\} \in E, y \in Y\}$, as desired.

Conversely, suppose the three conditions hold. For each $x \in X$, let $\min_x$ denote the least-cost edge between x and any vertex in Y it is adjacent to. $\min_x$ is well-defined by condition (2). Note that computing the value

of $\min_x$ can be done in polynomial time. Since Y is connected, find a minimum spanning tree T_Y in the subgraph Y . By condition (3), we are guaranteed that the weight of this spanning tree T_Y of subgraph Y is at most $k - \sum_{x \in X} \min\{wt(\{x, y\}): \{x, y\} \in E, y \in Y\}$. Note that this step can also be done in polynomial time. To obtain a spanning tree T of G , for each $x \in X$, connect x to T_Y using an edge of cost $\min_x$. Clearly, the spanning tree T of G constructed has the property that $wt(T) \leq k$ and $X \subseteq \Pi_T$. $\square$

Since conditions (1), (2) and (3) of Theorem 3.3 can be determined in polynomial time, we have the following result.

Corollary 3.4. Problem 3.2 can be solved in polynomial time.

4. Conclusion

In this paper, we were able to show that a variation of the maximum leaf spanning tree problem, for bipartite graphs, is NP-complete. In addition, it remains NP-complete for planar bipartite graphs of maximum degree 4.

We have also generalized a problem that is polynomial-time solvable to weighted-graphs and shown that this problem is still polynomial-time solvable.

Finally, it is known that the maximum leaf spanning tree problem remains NP-complete for cubic graphs [3]. It would be nice to be able to determine whether Problem 1.2 remains NP-complete for cubic bipartite graphs.

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